Relevance-Guided Adaptive Learning for Remote Sensing Image–Text Retrieval

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***Abstract*— The remote sensing image–text retrieval (RSITR) aims to establish semantic alignment between images and texts to enable accurate cross-modal retrieval. Existing methods usually extract features from images and texts independently, aligning them in a shared embedding space to achieve cross-modal retrieval. However, these methods often assume complete alignment between image and text pairs, overlooking the inherent disparities between the rich visual details in remote sensing (RS) images and the abstract nature of textual descriptions. These disparities result in image–text pairs only sharing partial semantic correlations, rather than one-to-one complete alignment. Such incomplete alignment adversely affects model training and retrieval accuracy. To address this problem, a relevance-guided adaptive learning (RGAL) method is proposed, which quantifies and leverages the relevance of image–text pairs to refine the training process while enhancing retrieval performance. First, the proposed method introduces an image–text relevance measurement mechanism that integrates global and local feature distances to accurately evaluate the degree of semantic relevance between images and texts. Second, a relevance-based sample division (RBSD) strategy is proposed, utilizing a Gaussian mixture model to dynamically redivide samples into positive and negative pairs according to the measured image–text relevance. This strategy refines the training dataset, reduces noise, and enhances the effectiveness of model learning. Finally, a relevance-weighted triplet loss (RWTL) is designed to adaptively adjust the contribution of sample pairs to the loss function based on their relevance, further optimizing model training and enhancing retrieval accuracy. Experimental results on multiple RSITR datasets demonstrate that the proposed method significantly improves retrieval accuracy and performance.**

***Index Terms*— Incomplete alignment, relevance-based sample division (RBSD), relevance-weighted triplet loss (RWTL), remote sensing image–text retrieval (RSITR), semantic alignment.**

I. Introduction

REMOTE sensing image–text retrieval (RSITR) is a cross-modal retrieval task that leverages the relationship between remote sensing (RS) images and textual information. Specifically, it involves retrieving the most relevant image or text given an input image or text [1]. RSITR can rapidly obtain relevant information, playing a pivotal role in applications, such as RS monitoring, environmental protection, and disaster assessment [2], [3]. RS images and texts are inherently distinct in their characteristics. While images capture complex spatial and spectral features, texts convey higher level semantic information through linguistic descriptions. These differences result in substantial discrepancies in their expression modes, structures, and semantics [4], [5]. However, the success of retrieval hinges on the proper alignment of the cross-modal features. As a result, the accurate alignment of image and text features is a fundamental challenge in RSITR.

Existing methods usually use image and text encoders to extract features, respectively, and then use constraint information to train the model for cross-modal feature alignment [6], [7], [8]. Chen et al. [6] designed a clue-driven multiassociation feature matching network for RSITR, which enhances the image–text feature alignment by incorporating both fine-grained and coarse-grained clues. He et al. [9] proposed an end-to-end visual–language fusion framework that combines separate visual and language encoders with a multimodal encoder, enhancing cross-modal feature alignment for retrieval. In recent years, deep learning techniques such as cross-modal attention mechanisms have promoted the progress of cross-modal feature alignment [10], [11]. Yuan et al. [10] developed a multiscale vision attention module to extract key features from RS images and guide text representation, enhancing cross-modal alignment and retrieval performance. Zhang et al. [12] proposed a mask-guided relation attention modeling method to improve retrieval accuracy by focusing on entity and relation features.

Although existing RSITR methods have made progress, they often focus on extracting and associating regional features of RS images and texts to achieve cross-modal alignment. However, they tend to overlook the issue of incomplete alignment between RS images and texts. Unlike natural images, RS imagery poses unique challenges due to scale variability, complex spatial composition, and subtle interclass differences. Moreover, textual descriptions associated with RS images are typically short and abstract, often focusing on dominant land cover types while neglecting other salient details. As a result, image–text alignment in RS is often incomplete and noisy. Fig. 1 illustrates the incomplete alignment between RS images and texts. For instance, the first image contains rich visual details such as green areas, building forms, and orientations within a residential area. However, the corresponding texts

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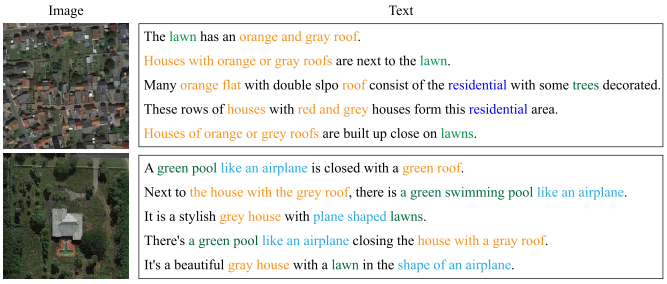


Fig. 1. Examples illustrating incomplete alignment between RS images and their textual descriptions. Texts of the same color correspond to the same entity in the image. 1) Textual descriptions often capture only a subset of visual content within the image. For instance, green text describes vegetation features while omitting surrounding structures; orange text details buildings but fails to mention orientations or adjacent roads. 2) Different descriptions of the same image exhibit significant variation in focus and terminology. For instance, both "lawn" and "pool" are used to describe the plane-shaped object in the second example.

only provide generalized descriptions like "houses," "flat," and "orange or gray roofs," lacking sufficient detailed information. The second image shows a plane-shaped object ambiguously labeled as "lawn" or "pool." These examples demonstrate that the relationship between image and text involves partial matches rather than precise one-to-one correspondence.

Although recent efforts have explored feature fusion, semantic enhancement, and contrastive learning to improve the RSITR performance, few works have directly addressed the problem of incomplete alignment. In real-world settings, not all image–text pairs contribute equally to model training. Some are well-aligned while others are only loosely or partially matched. Treating all samples uniformly can introduce noise and degrade retrieval performance.

To tackle this challenge, this article proposes a relevance-guided adaptive learning (RGAL) method for RSITR. The core idea is to assess the semantic relevance of each image–text pair and leverage this information to guide both sample selection and loss weighting during training. Specifically, the proposed method computes pairwise relevance using both global and local feature distances, performs sample division via Gaussian mixture model (GMM), and applies a relevance-weighted triplet loss (RWTL) to emphasize ambiguous or informative examples. The overall framework of the proposed method is shown in Fig. 2. It consists of three main components: image–text relevance measurement, relevance-based sample division (RBSD), and RWTL. First, in the image–text relevance measurement, both global and local features of images and texts are extracted. Global features capture overall semantic information, while local features focus on specific details or regions. The semantic relevance between images and texts is then quantified by calculating the distance between corresponding cross-modal features. Second, in the RBSD, the relevance distribution of image–text pairs is modeled using a GMM. Based on the quantified relevance, GMM is used to divide the pairs into high-relevance positive samples and low-relevance negative samples, providing more reliable training data for the model. Finally, an RWTL is introduced to optimize the model using the refined dataset. The relevance of image–text pairs is incorporated as a coefficient in the loss term. As a result, the model adaptively focuses more on positive pairs with lower relevance and negative pairs with higher relevance during training, improving its ability to learn discriminative cross-modal features and enhancing overall retrieval performance.

The effectiveness of the proposed method is validated through experiments on two public RSITR datasets: RSICD [13] and RSITMD [10]. The results indicate superior robustness and accuracy, providing new insights for improving RSITR performance. The main contributions of this article are summarized as follows.

1) A novel RGAL method is introduced to explicitly model and exploit semantic relevance, addressing the incomplete alignment issue in RSITR.
2) An RBSD strategy is designed to improve the quality of training data and reduce noise interference.
3) An RWTL is proposed to adaptively adjust the contribution of each sample pair based on its relevance, optimizing the model training.

II. Related Works

A. Remote Sensing Image–Text Retrieval

The RSITR task aims to establish semantic alignment between RS images and texts, enabling retrieval of relevant images based on textual descriptions or retrieval of corresponding texts based on image query [14], [15], [16]. In recent years, the widespread use of deep learning technologies, particularly convolutional neural networks (CNNs) and transformer architectures, has significantly advanced this field [17], [18], [19].

Methods based on deep CNNs (DCNNs) exploit DCNNs' strength in image feature extraction and recurrent neural networks (RNNs) ability to process text sequences [20], [21], [22]. These methods use fusion strategies to link image and text features. Abdullah et al. [21] introduced a deep bidirectional triplet network for text-to-image (T2I) matching to enhance alignment between textual and visual representations. Zhang et al. [22] proposed an adaptive alignment strategy with curriculum learning to progressively align image–text pairs. Ji et al. [23] used the momentum contrastive learning to build a dynamic dictionary with diverse negative samples, improving semantic differentiation. However, due to DCNNs' inductive bias, capturing long-range dependencies and global semantic associations in complex RS scenes remains challenging. To address this, some studies incorporate multiscale feature fusion and adaptive attention mechanisms [24], [25]. For example, Yuan et al. [24] refined global features with local information and enhanced local details with global context, dynamically integrating both to generate more salient visual representations.

Transformer-based methods use the self-attention mechanism to process both image and text data simultaneously [26], [27], capturing long-range dependencies and global information to better explore semantic associations. Recent advances in RSITR have been driven by contrastive language-image pre-training (CLIP) [2], [28], which learns semantic alignment between images and texts through large-scale pretraining [29]. Rahhal et al. [30] used vision and language

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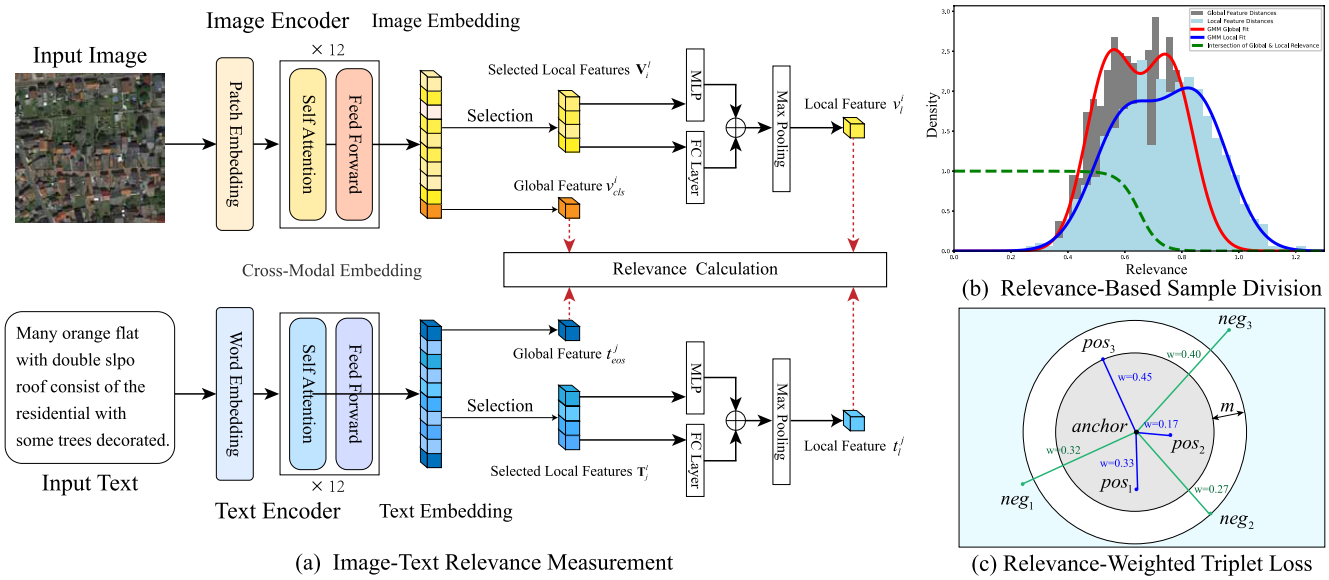


Fig. 2. Illustration of the proposed RGAL method. (a) Global and local features of images and texts are extracted through a dual-branch network, and the image–text relevance is calculated based on the cross-modal feature distance. (b) GMM is applied to redivide image–text pairs based on their relevance, ensuring high-quality training data. (c) Relevance of image–text pairs is incorporated into the loss function, adaptively adjusting sample contributions to optimize the model training.

Transformers for cross-modal retrieval, later developing a multilingual framework for English, Arabic, French, and Italian queries [31]. Ma et al. [2] introduced DOVE, a model that enhances visual–text alignment with a region-oriented attention module. Yang et al. [26] proposed a multiscale cross-modal alignment Transformer to improve image–text alignment across multiple scales.

Existing methods often ignore the problem of imperfect alignment between RS images and text. This misalignment stems from the rich spatial details in the image and the abstract nature of the text description. To address this issue, this article proposes to measure the image–text correlation through feature distance calculation and optimize the training data based on the measured correlation to enhance model training.

B. Image–Text Semantic Alignment

Semantic alignment allows the calculation of similarity between images and texts within a shared feature space, which is essential for effective image–text retrieval [32], [33]. The semantic alignment typically involves two approaches: global alignment and local alignment.

Global alignment focuses on matching image and text features from a global perspective, which often uses a two-stream network to learn global features and calculate the similarity between them [34], [35]. In recent years, pretraining and fine-tuning methods based on global alignment have yielded promising results [36]. These methods enhance the representation of global features by leveraging extensive datasets during pretraining. For example, Gao et al. [37] enhanced image–text alignment in open-vocabulary settings, which improved downstream tasks such as image classification, retrieval, and generation. However, global alignment methods may fail to capture detailed features, leading to inaccurate similarity calculations between image–text pairs. To address this limitation, introducing local alignment as a supplementary solution is crucial.

Local alignment emphasizes the correspondence between specific regions in images and individual words in texts [38], [39], [40]. A common approach is to employ attention mechanisms to identify semantic relationships between image regions and text components [41], [42], [43]. For example, Chen et al. [42] designed an adaptive local component selection method based on an attention mechanism. Zhang et al. [44] proposed a context-aware attention method that adaptively selects local elements guided by global information to enhance alignment accuracy. Wu et al. [45] introduced a spatial–channel Transformer with pseudo-regions to enhance local representation.

Global and local alignment play complementary roles in facilitating accurate image–text retrieval. In this article, the image–text relevance is measured by jointly considering global and local feature distances, enabling a more comprehensive and precise representation of cross-modal semantic associations.

C. Vision–Language Pretraining

Recent advances in vision–language pretrained models (VLMs), such as CLIP [29], BLIP [46], and Flamingo [47], have significantly improved cross-modal understanding in natural image domains. These models leverage large-scale pretraining on web-scale data, enabling impressive zero-shot and few-shot performance [7], [48]. However, their direct application to RS tasks such as RSITR remains nontrivial [2]. This work builds upon CLIP, a representative vision–language model pretrained on large-scale natural image–text pairs. While CLIP provides strong general-purpose representations, directly applying it to RS tasks remains challenging due to domain discrepancies. RS images differ significantly from natural images in terms of geometry, spectral range,

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and semantic distribution. In addition, CLIP and similar VLMs do not explicitly handle the partial or incomplete alignment often found in RS image–text pairs. To address these limitations, the proposed RGAL method integrates CLIP encoders as the backbone but introduces a relevance-guided training strategy that adapts to semantic inconsistencies. This design improves the model's robustness in RS scenarios without requiring large-scale domain-specific pretraining. Future work may explore incorporating more advanced VLMs or domain-adaptive pretraining strategies to further enhance performance.

III. Proposed Method

This section discusses the proposed RGAL in detail. RS images often contain rich visual details, while text descriptions typically focus on a limited subset of these elements. This mismatch leads to incomplete alignment between image–text pairs. To address this, the proposed method first measures how semantically relevant each image–text pair is by comparing their global and local features (see Section III-A). Based on this relevance, the model dynamically adjusts the training samples (see Section III-B) and the loss function (see Section III-C) to focus more on ambiguous or hard-to-learn examples. This process enables the model to better learn cross-modal relationships, even when the data are partially misaligned.

A. Image–Text Relevance Measurement

In multimodal learning, images and texts are typically mapped to a shared semantic space using feature extractors, aiming to bring similar images and texts closer while distancing dissimilar ones [31], [49]. Therefore, this article introduces an image–text relevance measurement that quantifies their semantic relevance by calculating the cross-modal feature distance between them. Following previous works [26], [50], the image and text encoders of the pretrained CLIP [29] model are exploited to extract both global and local features from images and texts.

1) Image Feature Extraction: The input image I_i is divided into N_* fixed size, nonoverlapping image patches. Each patch is treated as a local area of the image. To effectively extract global features, a [CLS] token is introduced to serve as the global feature identifier for the image. Both the image patches and the [CLS] token are input into the image encoder to extract image features

$$\mathbf{V}_i = \left\{v^i_{\text{cls}}, v^i_1, v^i_2, \ldots, v^i_{N_*}\right\} \in \mathbb{R}^{(N_*+1)\times d} \tag{1}$$

where d is the dimension of the latent feature space, v^i_{cls} represents the global feature of the image I_i, and each v^i_j reflects the characteristics of the individual local regions.

2) Text Feature Extraction: For an input text T_i, the text is first converted into a sequence of symbols. Each word in the text is mapped to an embedding vector using a lower cased byte pair encoding (BPE) method [51] with a vocabulary size of 49 152. Before inputting the text sequence into the text encoder, the start symbol [SOS] and the end symbol [EOS] are added to both ends of the sequence. This text sequence is then input into the text encoder to extract text features

$$\mathbf{T}_i = \left\{t^i_{\text{sos}}, t^i_1, t^i_2, \ldots, t^i_{N_o}, t^i_{\text{eos}}\right\} \in \mathbb{R}^{(N_o+2)\times d} \tag{2}$$

where N_o indicates the length of the input text, t^i_j denotes the feature of each word, and t^i_{eos} serves as the global feature representing the overall meaning of the text T_i.

3) Cross-Modal Feature Distance: For an image–text pair (I_i, T_j), the global feature distance is defined as the cosine similarity between the global embedding representations of the image and the text

$$d^g_{ij} = \text{sim}\left(v^i_{\text{cls}}, t^j_{\text{eos}}\right) = \frac{v^i_{\text{cls}} \cdot t^j_{\text{eos}}}{\left\|v^i_{\text{cls}}\right\|\left\|t^j_{\text{eos}}\right\|}. \tag{3}$$

Relying solely on global similarity may overlook subtle but significant differences between images and text. Therefore, the local feature distance is also considered. In order to effectively distinguish local features from global features, a certain proportion of local features is selected for aggregation. Specifically, according to the weight value of the last attention layer of the Transformer block in the feature extractor, the first R proportion of image patch features or word features with larger weights are selected. These selected local features are then aggregated through a multilayer perceptron (MLP), a fully connected (FC) layer, and a maximum pooling (MaxPool) operation to obtain the final local features. The process for obtaining the aggregated local feature of the input image can be mathematically expressed as follows:

$$v^i_l = \text{MaxPool}\left(\text{MLP}\left(\mathbf{V}^l_i\right) + \text{FC}\left(\mathbf{V}^l_i\right)\right) \tag{4}$$

where $\mathbf{V}^l_i \in \mathbb{R}^{\text{RN}_*\times d}$ represents the matrix composed of the selected local image block features. Similarly, the process to obtain the fine-grained text representation can be formulated as follows:

$$t^j_l = \text{MaxPool}\left(\text{MLP}\left(\mathbf{T}^l_j\right) + \text{FC}\left(\mathbf{T}^l_j\right)\right) \tag{5}$$

where $\mathbf{T}^l_j \in \mathbb{R}^{\text{RN}_o\times d}$ represents the matrix composed of the selected word features. The local feature distance for the image–text pair (I_i, T_j) is then defined as follows:

$$d^l_{ij} = \text{sim}\left(v^i_l, t^j_l\right) = \frac{v^i_l \cdot t^j_l}{\left\|v^i_l\right\|\left\|t^j_l\right\|}. \tag{6}$$

The proposed method measures the relevance of image–text pairs using cross-modal feature distances, including both d^g_{ij} and d^l_{ij}. Specifically, a larger distance between image and text indicates lower relevance, signifying a greater semantic difference. Conversely, a smaller distance indicates higher relevance, reflecting a stronger semantic association. This distance-based measurement approach effectively captures the matching degree between image and text, providing a solid foundation for accurate image–text relevance division and effective model training.

B. Relevance-Based Sample Division

Based on the measurement of image–text relevance, an RBSD strategy is proposed to reorganize the training data, ensuring the reliability of the dataset. Highly related image–text pairs are treated as positive samples, while less related pairs are considered negative samples. This division reduces the impact of noise and improves the quality of the

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training data, ultimately enhancing the model's ability to learn accurate cross-modal representations for RSITR. Specifically, a Gaussian mixture model is used to model the relevance of image–text pairs. The posterior probabilities are then calculated to classify the pairs into positive or negative sample based on their relevance.

1) Gaussian Mixture Modeling: The Gaussian mixture model is a probability-based approach that assumes data are a mixture of multiple Gaussian distributions. It effectively handles complex data distributions and uncertainties. Compared with K-means or density-based clustering algorithms such as DBSCAN, GMM is more suitable for modeling image–text correlation. On one hand, GMM supports soft assignment, avoiding the errors caused by hard division; on the other hand, its probabilistic modeling framework can fit the feature distance distribution more flexibly and better handle the ambiguity and overlapping areas between samples. In this study, a two-component Gaussian mixture model is used to fit the relevance distribution of image–text pairs for both positive and negative samples. This modeling assumption allows for a more accurate capture of the relevance differences between images and texts. Specifically, GMM learns two Gaussian distributions by fitting the feature distance vectors from the training set before each training epoch, categorizing the pairs into high-relevance positive samples and low-relevance negative samples. The expectation–maximization algorithm is used to train the GMM, yielding optimal model parameters

$$p(x) = \pi_1 \mathcal{N}\left(x|\mu_1, \sigma_1^2\right) + \pi_2 \mathcal{N}\left(x|\mu_2, \sigma_2^2\right) \tag{7}$$

where π_1 and π_2 are the weights of the two Gaussian components, satisfying $\pi_1 + \pi_2 = 1$, μ_1 and μ_2 are the means of the two components, and σ_1^2 and σ_2^2 are the variances of the two components.

2) Division Based on the Posterior Probability: After obtaining the Gaussian mixture model, the feature distance vector of each image–text pair is assigned to one of the two categories. For example, considering the global feature distance, the posterior probability of each image–text pair belonging to each category is calculated as follows:

$$P\left(C_k \mid d_{ij}^g\right) = \frac{P\left(d_{ij}^g \mid \mu_k, \sigma_k^2\right) P(C_k)}{\sum_{k'} P\left(d_{ij}^g \mid \mu_{k'}, \sigma_{k'}^2\right) P(C_{k'})} \tag{8}$$

where $P(C_k)$ is the prior probability of category C_k, $C_k \in \{0, 1\}$ indicates whether an image–text pair belongs to the positive sample set (where $C_k = 1$) or the negative sample set (where $C_k = 0$), and $P(d_{ij}^g \mid \mu_k, \sigma_k^2)$ is the likelihood of d_{ij}^g in category C_k. Based on the maximum posterior probability, the image–text pair (I_i, T_j) is assigned to the most likely category.

According to the global feature distance modeling, the positive sample set I_i^{g+} and the negative sample set I_i^{g-} for image I_i in a training batch $\mathcal{B}$ are defined as follows:

$$\begin{aligned} I_i^{g+} &= \left\{T_j \mid P\left(C_k = 1 \mid d_{ij}^g\right) \geq \tau \quad \forall T_j \in \mathcal{B}\right\} \\ I_i^{g-} &= \left\{T_j \mid P\left(C_k = 1 \mid d_{ij}^g\right) < \tau \quad \forall T_j \in \mathcal{B}\right\} \end{aligned} \tag{9}$$

where τ is the threshold for division. The positive sample set I_i^{g+} contains texts that have a strong semantic match with the image I_i. Conversely, the negative sample set I_i^{g-} contains texts that have a weak semantic relationship.

The positive sample set I_i^{l+} and the negative sample set I_i^{l-} based on the local feature distance can be obtained in a similar way. The final sets of positive and negative sample pairs are determined by the intersection of the global and local division results

$$\begin{aligned} I_i^+ &= I_i^{g+} \cap I_i^{l+} \\ I_i^- &= I_i^{g-} \cap I_i^{l-}. \end{aligned} \tag{10}$$

Similarly, the cross-modal positive sample set T_i^+ and negative sample set T_i^- of the text T_i can also be obtained. This fine-grained division allows the model to obtain higher quality training data, thus improving the performance of image–text retrieval tasks.

C. Relevance-Weighted Triplet Loss

The traditional triplet loss function primarily focuses on the distance between the anchor and a single positive or negative sample. This limitation makes the model vulnerable to noise interference during training, affecting both learning performance and effectiveness. To address this issue, the proposed method introduces an RWTL. RWTL considers all sample pairs within the training batch and adjusts their influence on model training by introducing a weighted coefficient based on the measured image–text relevance. Positive sample pairs with low relevance and negative sample pairs with high relevance contribute a larger loss value. This enables the model to focus more on difficult, hard-to-distinguish samples during training, improving the alignment of decision boundaries and enhancing the model's ability to learn discriminative cross-modal features. Moreover, the weighting mechanism helps to prevent overfitting on easily distinguishable samples, thus improving the stability and robustness of the training process.

The traditional triplet loss function can be formulated as follows:

$$\mathcal{L}_{\text{tri}}(\mathbf{a}) = \left[d\left(\mathbf{a}, \mathbf{a}^+(i)\right) - d\left(\mathbf{a}, \mathbf{a}^-(i)\right) + m\right]_+ \tag{11}$$

where $[\cdot]_+ = \max(\cdot, 0)$, $d(\cdot, \cdot)$ denotes the cosine distance between the inputs, $\mathbf{a}^+(i)$ is a random positive sample associated with the anchor point $\mathbf{a}$, and $\mathbf{a}^-(i)$ is a random negative sample. m is a margin constant that ensures the distance of negative samples is greater than that of positive samples.

In the proposed RWTL, the distances between all positive and negative sample pairs in a batch are considered. As shown in Fig. 2(c), each sample pair is differentiated using a weight coefficient, which more accurately controls the impact of the sample pairs on the training process. The expression of the RWTL function is as follows:

$$\mathcal{L}_{\text{w-tri}}(\mathbf{a}) = \left[\Sigma_i \alpha_i d\left(\mathbf{a}, \mathbf{a}^+(i)\right) - \Sigma_j \beta_j \cdot d\left(\mathbf{a}, \mathbf{a}^-(j)\right) + m\right]_+ \tag{12}$$

where

$$\alpha_i = \frac{\exp\left(d\left(\mathbf{a}, \mathbf{a}^+(i)\right)/\epsilon\right)}{\sum_i \exp(d(\mathbf{a}, \mathbf{a}^+(i))/\epsilon)} \tag{13}$$

$$\beta_j = \frac{\exp\left(d\left(\mathbf{a}, \mathbf{a}^-(j)\right)/\epsilon\right)}{\sum_j \exp(d(\mathbf{a}, \mathbf{a}^-(j))/\epsilon).} \tag{14}$$

Here, α_i is the relevance weight of each positive sample, β_j is the relevance weight of each negative sample, $\exp(x) = e^x$,

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and ϵ is a temperature hyperparameter that regulates the sharpness of the probability distribution.

When training RSITR using RWTL, positive and negative sample sets are constructed for each image and each text in a training batch according to (10). By substituting the image I_i and text T_i into (12), the final loss function is obtained

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathcal{L}_{w\text{-tri}}(I_i)_g + \mathcal{L}_{w\text{-tri}}(I_i)_l + \mathcal{L}_{w\text{-tri}}(T_i)_g + \mathcal{L}_{w\text{-tri}}(T_i)_l\right) \tag{15}$$

where the subscripts g and l indicate that the corresponding loss terms are calculated based on the global and local feature distances, respectively, and N is the number of image–text pairs in the training batch $\mathcal{B}$. By incorporating RWTL and independently optimizing global and local features, the model enhances the alignment between images and texts, thereby improving RSITR performance.

IV. Experiments

A. Datasets and Metrics

Following prior work [24], [52], [53], the proposed method is evaluated on two standard RSITR datasets. The evaluation covers two key tasks: T2I retrieval and image-to-text (I2T) retrieval. In the T2I retrieval task, the objective is to retrieve the most relevant image from the image database based on a given textual query. Conversely, in the I2T task, the model is required to retrieve the corresponding descriptive text from the text library given an image query.

RSICD dataset [13] contains 10 921 high-resolution RS images from Google Earth, each paired with five captions, resulting in 24 333 total sentences. Due to repeated captions in most images, only 724 images have unique descriptions. Following previous methods [10], [24], the dataset is divided into 7862 training, 1966 validation, and 1093 testing images.

RSITMD dataset [10] includes 4743 images of 256 × 256 pixels, sourced from RSICD and Google Earth. Each image has five fine-grained captions focusing on detailed object relations. This dataset is split into 3435 training, 856 validation, and 452 testing images.

UCMerced-LandUse-Captions dataset [54], [55] contains 2100 aerial images across 21 land use categories, such as forest, residential, and airport. Each image has a resolution of 256 × 256 pixels with a spatial resolution of 0.3048 m. This dataset is split into 1680 training, 210 validation, and 210 testing images.

Sydney-Captions dataset [54], [56] is built upon high-resolution satellite imagery of Sydney, Australia, with an overall size of 18 000 × 14 000 pixels and a spatial resolution of 0.5 m. This dataset is split into 497 training, 58 validation, and 58 testing images.

Metrics: Following prior studies [3], [27], [50], $R@k$ and mR are adopted as evaluation metrics. $R@k$ measures the proportion of correctly retrieved results within the top k ranks, with higher values indicating better retrieval precision. This paper reports $R@1$, $R@5$, and $R@10$ for both T2I and I2T tasks. The mR metric, defined as the average of six $R@k$ scores, provides a more comprehensive evaluation by reflecting retrieval accuracy across multiple ranking depths. A higher mR score indicates robust performance over different retrieval levels.

B. Experimental Configurations

The proposed method was implemented in PyTorch and trained on an NVIDIA GeForce RTX 3090 GPU. For initialization, the image and text encoders leverage pretrained CLIP parameters to enhance the feature extraction efficiency. Input images were resized to 256 × 256, and data augmentation techniques such as random flipping, padded cropping, and erasing were applied. The maximum text sequence length was set to 77. The Adam optimizer was employed for 60 epochs. The initial learning rates were 1×10^{-5} for CLIP parameters and 1×10^{-3} for local feature extraction. A cosine decay schedule was used, with the learning rate gradually increasing from 1×10^{-6} to 1×10^{-5} over the first five warm-up epochs. In addition, the margin value m of the triplet loss was set to 0.1, the temperature parameter ϵ was fixed at 0.02, the division threshold τ was set to 0.7, and the selection ratio R of the local features was set to 0.3. The GMM was refit at the beginning of each training epoch using the normalized loss values computed for each sample. For the GMM configuration, the maximum number of iterations was set to 10 to ensure computational efficiency. The convergence tolerance was set to 0.01, allowing early termination when the log-likelihood improvement became negligible. A regularization term of 5×10^{-4} was added to the covariance matrix to maintain numerical stability.

C. Ablation Study

To thoroughly evaluate the effectiveness of the proposed method, two sets of ablation experiments were conducted. The first set examined the independent contribution of each component module. The second set investigated the role of global and local features in measuring the relevance between images and texts.

1) Ablation Analysis of Method Components: This section verifies the effectiveness of each component in the proposed method. The experimental results are presented in Table I. Ablation experiments compare the performance of four settings: the baseline model (baseline), the model with only RBSD, the model with only RWTL, and the proposed complete method with both RBSD and RWTL (RGAL). A detailed analysis of each setting follows.

The experimental results showed that the baseline model achieved 7.96% T2I $R@1$ and 20.90% mR on the RSICD dataset, and 10.84% T2I $R@1$ and 27.42% mR on the RSITMD dataset. These results were significantly worse than those of other configurations with optimization strategies. The baseline model used a dual-branch structure to extract image and text features, followed by multimodal embedding for alignment under the traditional triplet loss constraint. In this configuration, the model did not consider the incomplete alignment between RS images and texts.

In the second RBSD setting, RBSD was added to the baseline model. Compared with the baseline model, RBSD showed improvements across all indicators. Specifically, the

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TABLE I
ABLATION EXPERIMENTAL RESULTS FOR COMPONENT EVALUATION ON RSICD AND RSITMD DATASETS

	Methods	RSICD dataset							RSITMD dataset						
		T2I			I2T			mR	T2I			I2T			mR
		R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
1	baseline	7.96	22.69	34.36	6.11	21.23	33.03	20.90	10.84	31.73	48.67	9.34	27.43	36.50	27.42
2	RBSD	8.20	29.42	45.22	7.35	27.26	37.80	25.88	15.83	35.96	49.77	13.54	32.44	41.63	31.53
3	RWTL	17.26	34.07	49.12	14.38	36.73	45.35	32.82	27.56	47.39	65.80	25.44	44.25	64.47	45.82
4	RGAL	19.58	41.73	54.64	16.44	40.58	52.52	37.58	29.43	53.50	69.25	27.43	56.24	70.97	51.14

mR score increased by +4.98% and +4.11% on the RSICD and RSITMD datasets, respectively. In addition, indicators such as $R@1$, $R@5$, and $R@10$ exhibited stable improvements. RBSD achieved this by combining global and local feature distances to redivide image–text pairs, thereby improving the quality of training data and strengthening alignment.

In the third RWTL setting, the RWTL was adopted based on the baseline model. RWTL achieved significant performance improvements compared with the baseline model. Specifically, the mR index increased by +11.92% and +18.4% on the two datasets, respectively. The traditional triplet loss only considered random positive and negative sample pairs. In contrast, RWTL considered all positive and negative sample pairs in a training batch. It performed differentiated processing based on feature distance, allowing the model to better adjust the weights of each sample pair. This optimization helped to refine the decision boundary and improve retrieval accuracy.

Finally, RBSD and RWTL were combined to form the complete RGAL method, further enhancing model performance. Specifically, the RGAL method achieved 19.58% T2I $R@1$ and 37.58% mR on the RSICD dataset, and 29.43% T2I $R@1$ and 51.14% mR on the RSITMD dataset, outperforming all other settings. The results showed that while RBSD or RWTL improved performance when used individually, the combination of both yielded superior results. RBSD enhanced model training by optimizing the training data based on image–text relevance. RWTL improved the training process and refined the decision boundary by adjusting the weights of positive and negative sample pairs differentially. The synergy of these two components enabled the RGAL method to excel in both global and local semantic alignment, significantly improving the RSITR performance.

2) Ablation Analysis of Global and Local Feature Contributions: This section analyzes the impact of global and local features on image–text relevance measurement. It compares the performance of three feature configurations. Specifically, the RGAL_g method uses only global features, the RGAL_l method uses only local features, and the RGAL_{g+l} method combines both global and local features. The experimental results are presented in Table II.

The RGAL_g method, using only global features, achieved 36.41% and 48.50% mR on the RSICD and RSITMD datasets, respectively. This indicated that global features capture the overall semantic information of images and texts to some extent. In contrast, the RGAL_l method, which used only local features, achieved mR scores of 34.66% and 47.61%, which were 1.75% and 0.89% lower than those of the RGAL_g method. This result suggested that while local features improved fine-grained semantic alignment by focusing on detailed semantics, the absence of global feature guidance limited its effectiveness in complex scenes. When global and local features were combined, the RGAL_{g+l} method achieved 37.58% and 57.14% mR on the RSICD and RSITMD datasets, respectively. These improvements of 1.17% and 8.64% over the RGAL_g method, and 2.92% and 9.53% over the RGAL_l method, demonstrated the significant complementarity between global and local features in semantic alignment. Global features provided macrolevel guidance, while local features enhanced the capture of details. The effective fusion of both enabled the model to better measure the semantic relevance between images and texts, thereby significantly improving image–text retrieval performance.

TABLE II
PERFORMANCE ANALYSIS OF DIFFERENT FEATURE CONFIGURATIONS. THE mR METRIC ON THE RSICD AND RSITMD DATASETS IS SHOWN

Methods	RSICD dataset	RSITMD dataset
RGAL_g	36.41	48.50
RGAL_l	34.66	47.61
RGAL_{g+l}	37.58	51.14

D. Parameter Analysis

This section examines the influence of key parameters on model performance, specifically focusing on the effects of the local feature selection ratio R and the RBSD threshold τ on overall model performance.

1) Local Patch Selection Ratio R: The local feature selection ratio R determines the number of features selected during the local feature extraction process in Section III-A. Specifically, the model calculates the correlation between each local patch feature and the global feature, selecting the most relevant R ratio of local patch features. These selected features are then aggregated into a final local feature vector.

To explore the impact of R on retrieval performance, this section tests the proposed RGAL method on the RSICD and

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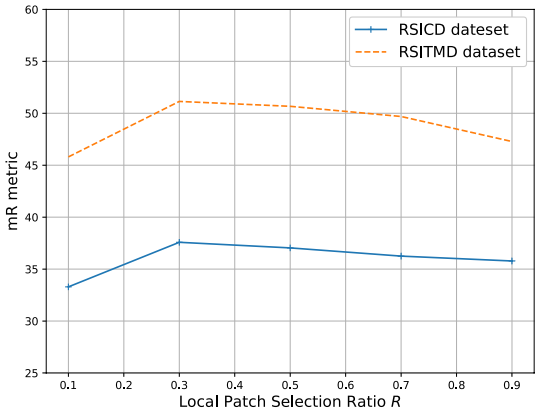


Fig. 3. Parameter analysis of local patch selection ratio R on the RSICD and RSITMD datasets.

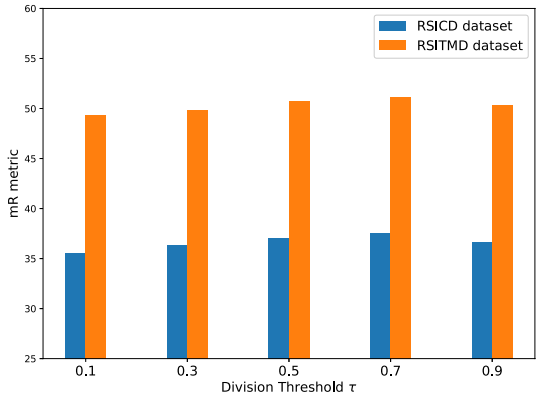


Fig. 4. Parameter analysis of division threshold τ on the RSICD and RSITMD datasets.

RSITMD datasets for values of $R = 0.1, 0.3, \ldots, 0.9$. The experimental results are shown in Fig. 3. As R increases, the model's performance exhibits a "rise and then fall" trend. This suggests that the local feature selection ratio plays a critical role in feature learning. Specifically, when $R = 0.3$, the model achieves the best performance. This indicates local features help compensate for the limitations of global features in describing complex scenes. However, when R exceeds 0.5, the redundancy in the local features increases, intensifying interference with the global features. This diminishes the feature interaction effect and negatively impacts overall model performance. These results suggest that a balance should be struck when selecting the proportion of local features to enhance semantic richness while avoiding information redundancy.

2) Relevance-Based Sample Redivision Threshold τ: The RBSD threshold τ is used to distinguish between positive and negative sample pairs. The image–text pair is classified as a positive or negative sample based on the maximum posterior probability of its relevance. If the probability exceeds the threshold τ, the pair is classified as positive; otherwise, it is classified as negative.

To evaluate the impact of τ on model performance, this section tests the proposed RGAL method with varying thresholds $\tau = 0.1, 0.3, \ldots, 0.9$ on two datasets. The experimental results are shown in Fig. 4. The results reveal that when $\tau < 0.5$, the model performance is lower than when $\tau \geq 0.5$. This result suggests that a threshold too low leads to incorrect classification of low-relevance image–text pairs as positive samples, reducing the model's ability to distinguish between positive and negative samples during training. When $\tau > 0.5$, performance gradually improves. This indicates that a higher threshold more accurately selects highly relevant image–text pairs, optimizing sample distribution and improving model training efficiency and generalization. The model performs best when $\tau = 0.7$. Increasing the threshold helps enhance the accuracy of sample division, reduces negative sample noise, and provides clearer targets for positive sample feature learning. However, a threshold too high (e.g., $\tau > 0.9$) may exclude some highly correlated image–text pairs, resulting in the loss of key information and a decrease in the model performance. Therefore, selecting the appropriate τ requires balancing the accuracy of sample division and the integrity of the data distribution.

E. Comparison With State-of-the-Art Methods

This section compares the performance of the proposed method with state-of-the-art methods on the RSICD and RSITMD datasets to assess its effectiveness and advantages. The comparison includes traditional deep learning methods and the latest Transformer-based methods. Traditional methods use CNNs for image feature extraction and RNNs for text feature extraction [24], [52], [53]. For instance, SWAN [53] improves scene awareness and reduces spatial semantic confusion by establishing correlations across features at different granularities. These methods emphasize key regions and features, improving retrieval accuracy. Transformer-based methods, such as CLIP, employ Transformer architectures to extract both image and text features [3], [6], [22]. For example, PE-RSITR [57] facilitates parameter-efficient transfer through the design of multi-modal adapters and contrastive loss functions.

1) Compared Results on RSICD Dataset: The experiments on the RSICD dataset are presented in Table III. The proposed method achieves competitive results across all evaluation metrics. Specifically, it obtains 37.58% mR, surpassing all compared methods. On the T2I task, the proposed method achieves an $R@1$ of 19.58%, outperforming the latest traditional method, KAMCL [23], by 7.59%, and surpassing the latest Transformer-based method, RemoteCLIP [58], by 4.85%. This demonstrates that the proposed method excels in both global and local semantic alignment of images and texts, leading to better accuracy in understanding text descriptions. On the I2T task, the proposed method achieves 16.44% $R@1$, outperforming all methods except RemoteCLIP [58]. The RemoteCLIP [58] method generates a pretraining dataset that is 12 times larger than all existing datasets combined by combining uncrewed aerial vehicle (UAV) imagery. This large-scale RS dataset improves the model to capture the diversity and complexity of images. Despite not using dedicated RS images for pretraining, the proposed RGAL method still achieves competitive results, highlighting its effectiveness. The success of the proposed

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TABLE III
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS ON THE RSICD DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	RSICD dataset						
	T2I			I2T			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
SIRS[52]	5.76	20.50	33.78	7.34	20.00	30.78	19.69
GaLR[24]	6.59	19.85	31.04	4.69	19.48	32.13	18.96
SWAN[53]	7.41	20.13	30.86	5.56	22.26	37.41	20.61
SMLGN[43]	8.87	25.53	37.24	7.85	27.14	42.58	24.87
SCAT-PRG[45]	10.88	26.32	39.66	8.06	25.43	40.31	25.11
KAMCL[23]	11.99	27.17	38.33	7.78	25.23	40.02	25.08
CDMAN[6]	7.13	25.38	40.35	10.23	24.94	36.83	24.14
HVSA[22]	7.47	20.62	32.11	5.51	21.13	34.13	20.16
VGSGN[3]	8.33	21.87	32.57	6.53	23.13	36.85	21.55
CUP[27]	9.86	31.60	47.15	12.05	32.66	48.70	30.34
MSA-CLIP[26]	10.16	25.71	36.96	7.87	25.67	41.85	24.70
PERSVL[50]	10.70	27.08	38.43	10.14	30.43	45.58	27.06
PE-RSITR[57]	14.13	31.51	44.78	11.63	33.92	50.73	31.12
RemoteCLIP[58]	14.73	39.93	**56.58**	**18.39**	37.42	51.05	36.35
RGAL	**19.58**	**41.73**	54.64	16.44	**40.58**	**52.52**	**37.58**

TABLE IV
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS ON RSITMD DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	RSITMD dataset						
	T2I			I2T			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
SIRS[52]	11.98	41.29	60.15	15.25	35.01	48.73	35.40
SWAN[53]	13.35	32.15	46.90	11.24	40.40	60.60	34.11
GaLR[24]	14.82	31.64	42.48	11.15	36.68	51.68	31.41
SCAT-PRG[45]	16.50	33.25	48.49	12.55	41.86	61.07	35.62
KAMCL[23]	16.81	34.96	47.12	14.60	41.86	59.86	35.87
SMLGN[43]	17.26	39.38	51.55	13.19	43.94	60.40	37.62
HVSA[22]	13.20	32.08	45.58	11.43	39.20	57.45	33.16
CDMAN[6]	13.82	41.96	59.74	16.46	37.74	51.24	36.83
VGSGN[3]	14.16	34.96	50.66	13.23	42.57	63.41	36.50
CUP[27]	17.70	48.40	64.82	21.57	43.70	56.75	42.16
PERSVL[50]	21.68	42.48	57.30	15.44	47.39	65.66	41.66
MSA-CLIP[26]	22.35	42.92	55.75	15.18	47.35	64.73	41.38
PE-RSITR[57]	23.67	44.07	60.36	20.10	50.63	67.97	44.47
RemoteCLIP[58]	23.76	**59.51**	**74.73**	**28.76**	52.43	63.94	50.52
RGAL	**29.43**	53.50	69.25	27.43	**56.24**	**70.97**	**51.14**

method is attributed to the RBSD mechanism, which improves training data quality. In addition, the RWTL enhances the association between image details and text semantics, further optimizing image–text alignment quality.

2) Compared Results on RSITMD Dataset: The experimental results on the RSITMD dataset are shown in Table IV. The proposed RGAL method achieved excellent performance in retrieval indicators such as T2I $R@1$ and mR, demonstrating its strong semantic matching ability for RS images and texts. Specifically, the method achieved 51.14% on mR, surpassing all the comparison methods, including RemoteCLIP [58]. This shows that the RGAL method can effectively capture the semantic alignment features of images and texts, thereby significantly improving the accuracy of image–text retrieval. It is worth noting that on the I2T task, the $R@1$ indicator of the RemoteCLIP method performs well, leading all the comparison methods. This is mainly due to the fact that RemoteCLIP uses a large-scale RS image dataset in the pretraining stage. In contrast, the proposed RGAL method still performs well without pretraining on large-scale RS datasets. This result suggests that image–text alignment does not only depends on the data scale but also the key lies in how to effectively perform semantic alignment. By introducing the RBSD and RWTL, the proposed RGAL method can accurately capture the image–text relationship, thereby maintaining strong competitiveness in complex RSITR tasks.

3) Compared Results on UCMerced-LandUse-Captions and Sydney-Captions Datasets: To further assess the generalization capability of the proposed method, additional experiments were conducted on the UCMerced-LandUse-Captions and Sydney-Captions datasets, following the same training and evaluation protocols as used on the RSICD and RSITMD datasets. These two datasets provide supplementary validation, despite being less commonly used in RSITR literature. Due to the lack of extensive prior benchmarks, comparisons with a limited set of representative methods are included.

As shown in Table V, the proposed RGAL method achieves the best overall performance on both datasets, demonstrating strong robustness and adaptability across diverse RS scenarios. It is worth noting that on the $R@10$ metric, RGAL slightly underperforms the interacting-enhancing feature transformer (IEFT) model [28]. This may be attributed to IEFTs more refined integration of global and local features, which better captures broader contextual cues and is particularly advantageous for higher recall thresholds such as $R@10$. Nonetheless, RGAL maintains competitive performance across all metrics. This observation also inspires future work to further enhance the RGAL method by incorporating more advanced feature interaction mechanisms.

F. Qualitative Experiments

This section presents qualitative examples to intuitively demonstrate the retrieval performance of the proposed method. It compares the results with those of the baseline model, covering both the I2Tand T2I tasks. The qualitative analysis highlights the performance differences across methods in practical applications, further verifying the effectiveness and robustness of the proposed method in the RSITR task.

1) Qualitative Experiments on T2I Task: The retrieval results are illustrated in Fig. 5. The proposed method outperforms the baseline model in terms of semantic consistency and detailed expression, particularly when handling text descriptions with complex keywords or detailed information. For

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TABLE V
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS ON UCMERCED-LANDUSE-CAPTIONS AND SYDNEY-CAPTIONS DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	UCMerced-LandUse-Captions dataset							Sydney-Captions dataset						
	T2I			I2T			mR	T2I			I2T			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
GaLR[24]	12.85	38.56	56.66	12.44	48.47	73.61	40.43	15.51	43.10	68.96	16.55	54.93	75.74	45.80
LW-WCR[18]	13.09	43.45	64.28	10.73	44.46	72.42	41.41	17.24	46.55	65.51	16.20	61.37	81.03	47.98
AMFMN[10]	16.23	50.39	68.64	13.58	49.81	75.71	45.73	18.96	53.44	70.68	20.68	58.96	**81.37**	50.68
IEFT[28]	16.50	57.94	82.54	15.75	60.56	**91.41**	54.12	21.84	**61.49**	72.99	20.45	60.23	78.81	52.64
RGAL	**17.62**	**58.86**	**87.24**	**18.10**	**61.10**	86.43	**54.89**	**22.07**	56.90	**78.28**	**24.14**	**63.45**	78.97	**53.97**

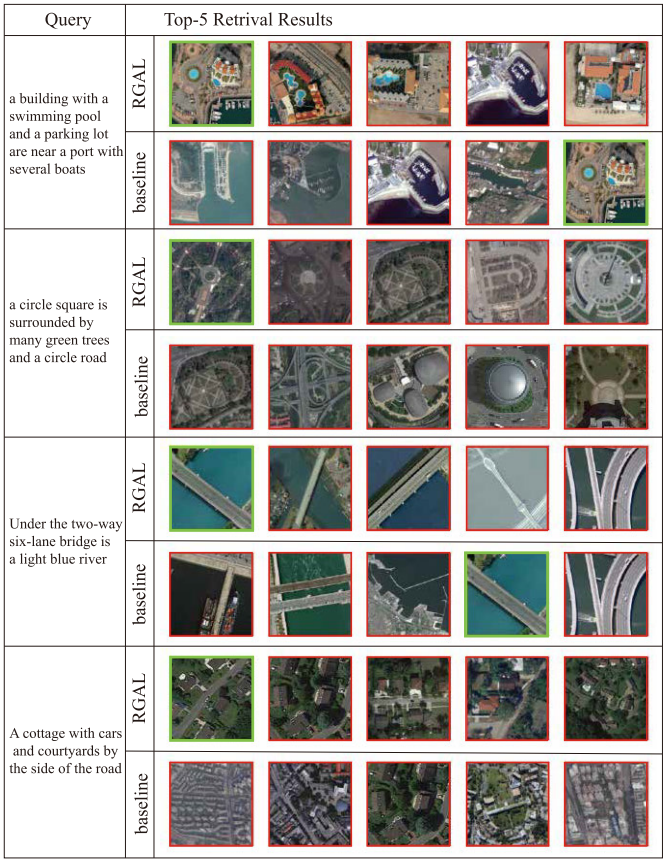


Fig. 5. Examples of retrieval results of the proposed RGAL method and the baseline model on the T2I task. Matching and nonmatching images are represented in green and red, respectively. The first two examples are from the RSICD dataset, and the last two examples are from the RSITMD dataset.



Fig. 6. Examples of retrieval results of the proposed RGAL method and the baseline model on the I2T task. Matching and nonmatching texts are represented in green and red, respectively. The first two examples are from the RSICD dataset, and the last two examples are from the RSITMD dataset.

example, with the input text "a building with a swimming pool and a parking lot near a port with several boats," the top-5 retrieval results of the proposed method all contain relevant elements such as "building" and "water." Four of the images depict buildings with swimming pools and parking lots. In contrast, the retrieval results of the baseline model include only one relevant image with a swimming pool, while the others return less relevant elements such as "port." Another example is the input text "a cottage with cars and courtyards by the side of the road." Both methods return images related to "cottage," but the proposed method more accurately selects images featuring "cars" and "courtyard" at the top results. The baseline model fails to prioritize these details, leading to a lower match between the images and text. The superior performance of the proposed RGAL method is attributed to the integration of RBSD and RWTL in the training process. These components enhance the model to focus on the semantic consistency between images and text, optimize the quality of training data, and significantly improve retrieval accuracy and robustness.

2) Qualitative Experiments on I2T Task: The experimental results are shown in Fig. 6. The proposed method significantly outperforms the baseline model in image–text matching accu-

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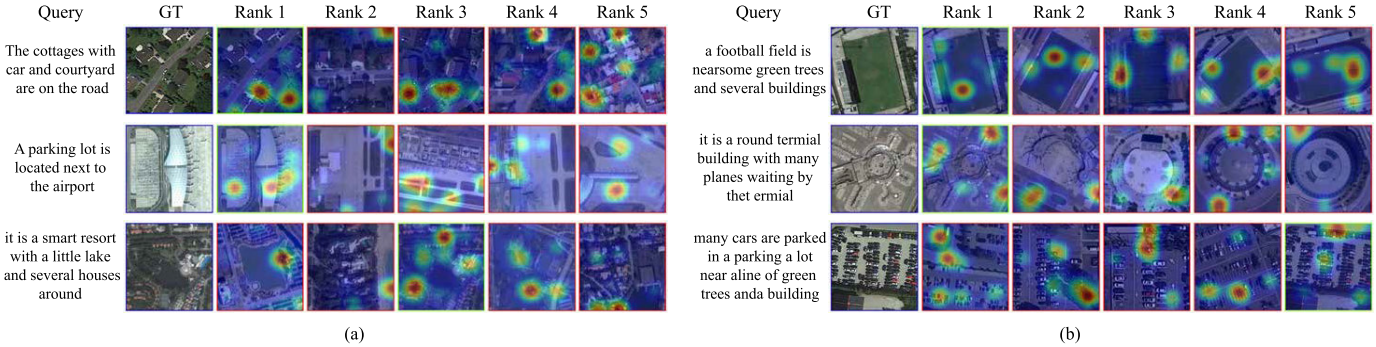


Fig. 7. Visualizations of attention-based heatmaps on two datasets. (a) RSITMD. (b) RSICD. The heatmaps highlight image regions that are most relevant to the query text in the T2I retrieval task. The first column presents the ground-truth image corresponding to the query text, while the second-to-sixth columns show the top-5 retrieved images ranked by the proposed RGAL method.

racy, especially in complex scenes. It captures fine-grained semantic information more accurately, enabling it to retrieve highly relevant text descriptions. For example, given an image of a house with a swimming pool in a forest, the proposed method retrieves all text descriptions that are completely relevant to the image. In contrast, while the baseline model retrieves keywords like "swimming pool," "house," and "building," it fails to describe the forest background and even incorrectly mentions the building color. In another example, for an image of a square building surrounded by a parking lot, the proposed method returns almost correct texts, such as "there is a majestic triangular center with a silver striped roof" and "a gray pavilion is in the center of the parking lot." While these descriptions contain some errors, they remain highly relevant to the overall scene. On the other hand, the baseline model returns incorrect results, which do not match the intuitive understanding of the image. For an image of school buildings with a football field, the proposed method retrieves texts which are highly relevant to the scene. Although the baseline model returns text containing the keyword "school," it provides insufficient details, such as missing references to the "football field" and failing to capture the key information.

In summary, the qualitative analysis on the I2T and T2I tasks clearly demonstrates the superior performance of the proposed method. Compared with the baseline model, the proposed method excels in detail capture, semantic consistency, and overall retrieval accuracy. Whether in I2T or T2I tasks, the proposed method provides more efficient and reliable retrieval capabilities and ensures accurate semantic alignment in complex scenarios.

3) Attention-Based Heatmap Visualization: To further assess the interpretability of the proposed RGAL method, attention-based heatmaps [59] are visualized under the T2I retrieval setting, as illustrated in Fig. 7. For each textual query, the top-5 retrieved images are presented with overlaid heatmaps that reflect the insights into which image regions the model attends to during cross-modal semantic alignment.

In the first example from the RSITMD dataset [first row in Fig. 7(a)], the query text refers to "the cottages with car and courtyard are on the road." The heatmaps across the retrieved images clearly show that the model focuses on regions containing cars and courtyard-like areas, often situated along roads. This behavior indicates that RGAL effectively captures multiple key objects and their spatial configurations as described in the query, thereby contributing to accurate retrieval.

In the third example from the RSICD dataset [third row in Fig. 7(b)], the query states "many cars are parked in a parking lot near a line of green trees and a building." The corresponding heatmaps prominently highlight areas within the retrieved images where clusters of vehicles are located, often adjacent to green vegetation or building structures. Although the ground-truth image is not ranked first, the model consistently attends to the semantically relevant regions described in the query. This demonstrates RGAL's robustness in identifying fine-grained semantic cues and its ability to retrieve contextually appropriate results. Future work may consider integrating more precise instance-level supervision to improve ranking precision.

These examples confirm that the proposed RGAL method consistently attends to image regions that are semantically aligned with the textual input, including salient objects and their spatial relationships. By successfully localizing meaningful content across diverse scenes in both the RSICD and RSITMD datasets, the model demonstrates strong generalization and interpretability. These observations further validate the effectiveness of the proposed RGAL method in addressing the incomplete alignment problem.

V. Conclusion

This article proposes an RGAL method to solve the problem of incomplete alignment between images and texts in RSITR. The proposed method introduces an image–text relevance measurement mechanism, an RBSD strategy, and a relevance-weighted triple loss to quantify and utilize the relevance of image–text pairs, optimize model training, and improve retrieval performance. Experimental results show that the proposed method has achieved significant performance improvements on two widely used RSITR datasets, demonstrating its effectiveness and robustness. While this work focuses on improving image–text alignment using a relevance-guided strategy within the CLIP framework, future research may explore the integration of more advanced VLMs. Future work may also focus on further refining the relevance measurement mechanism and integrating more precise instance-level supervision to enhance the model's understanding of RS images.

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